

Modeled Blood Glucose Imbalances induced by Hormonal Diseases

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1 Introduction

1.1 Background

1.1.1 Physiology

After consuming various forms of carbohydrate chains, the human body processes these to simple glucose to be either stored as chained glycogen or to be converted to cell usable energy through various downstream pathways. The intake of glucose into the bloodstream depends on the time, carbohydrate composition, blood glucose concentrations, and various bodily conditions. The cellular uptake of glucose out of the blood is mainly driven by the hormone insulin. An increase or decrease in production of the hormones leads to instability in holding stable blood glucose levels.

Properly diagnosing these imbalances allow for propermanagment of symptoms and options for intervention. [2]

1.1.2 Insulin Imbalances

Insulin Resistance: If insulin levels are considerably raised for a long period of time the cells of the body may develop a resistance to insulin. Insulin resistance requires a further increased insulin level for proper cellular action to lower blood sugar (hypoglycemia). Symptoms of insulin resistance are similar to type 1 diabetes but are usually with offset other hormonal interactions. If the resistance to the insulin is relieved but insulin production is not slowed then lowered blood sugar can lead to dizziness, shaking, confusion, and seizures.

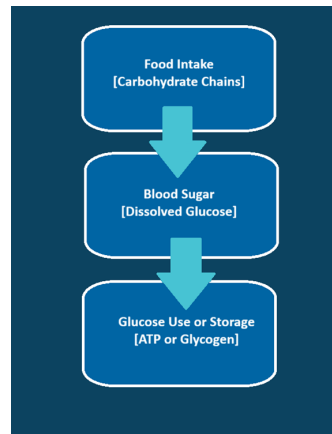


Figure 1: Blood Sugar Physiology

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Diabetes Type 1: The outcome of an inhibition of the pancreases' ability to produce adequate insulin due to autoimmune β -cell destruction. Type 1 diabetes results in lowered insulin levels leading to rising blood sugar levels hyperglycemia. Symptoms of type 1 diabetes elevated blood sugar include nausea, vomiting, irritability, dehydration, and fast heart rate (tachycardia). Eating large meals high in simple sugars without supplemental insulin to combat rising glucose levels can lead to dangerous sugar levels in seemingly healthy adults. Another cause of worry with type 1 diabetes is that blood sugar can also reach dangerously low levels if excessive supplemental insulin drives. [5]

Diabetes Type 2: The outcome of an inhibition of the pancreases' ability to produce adequate insulin due to non-autoimmune β -cell secretion inhibition. Type 2 diabetes is usually combined with and/or caused by cellular insulin resistance and exasperated by insulin resistance. Type 2 diabetes typically results in lower insulin levels for rising blood sugar levels and/or raised insulin levels to combat insulin resistance. Symptoms of type 2 diabetes are usually a combination of high and low blood sugar driven by interactions between diabetes and insulin resistance.

1.2 Model Assumptions

Some assumption of this model are:

- The human body uptakes glucose into the bloodstream through digestion.
- The hormone insulin promotes the cellular uptake of glucose for energy use.
- Insulin production is driven by glucose levels.
- Glucose can also be removed or added to the bloodstream for long-term storage.

This model also assumes that glucose and insulin blood concentrations can be modeled as first order non-linear differential equations. The change in blood glucose levels depend on:

- The input from digestion with time delay from consumption. This input is smoothed by a density function of absorbed glucose over time.
- The rate of storage removal or replenishment of glucose with respect to glucose concentrations.
- The rate of cellular uptake and consumption with respect to glucose and insulin levels.

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The change in blood insulin levels depend on:

- The rate of insulin production with respect to glucose levels.
- The rate of insulin removal with respect to insulin concentrations.

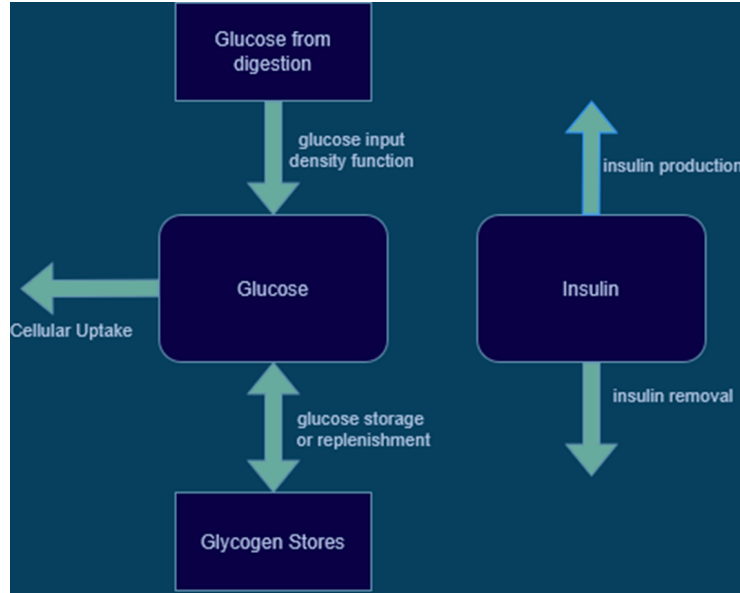


Figure 2: Model Structure

1.3 Hypothesis

This Model hopes to create a simple framework to be able to analyze wearable continuous glucose monitoring data to identify possible hormone imbalances. With this model this paper hopes to answer whether given an input glucose from digestion function what would be the responding glucose and insulin levels and what parameter conditions can both replicate and predict Type 1 and Type 2 Diabetes?

Hypothesis: Predictable outcomes from insulin driven diseases be given from expected changes in parameters.

2 Methods

2.1 Data

The data used for this model is taken from the HUPA-UCM diabetes dataset based on wearable continuous glucose monitors. [3]

2.2 Model Structure

This model is derived from Bridgewater et al, with input glucose replaced with a time dependent glucose density function. The model is defined by

$$\frac{dG}{dt} = f(t) - a_1 G - a_2 GI \quad (1)$$

$$(2)$$

$$\frac{dI}{dt} = b_1 G - b_2 I. \quad (3)$$

And with $f(t)$ defined by

$$f(t) = \frac{F}{\pi (1 + (t - \tau)^2)}$$

such that the total absorption over time equals F , i.e.

$$\int_{-\infty}^{\infty} f(t) dt = \int_{-\infty}^{\infty} \frac{F}{\pi (1 + (t - \tau)^2)} dt = F.$$

G	Blood Glucose Concentration $\left[\frac{\text{mg}}{\text{dl}}\right]$	
I	Blood Insulin Concentration $\left[\frac{\text{mIU}}{\text{dl}}\right]$	
t	time [min]	
a_1	Glucose removal and input rate not from Insulin $\left[\frac{1}{\text{min}}\right]$	
a_2	Glucose removal rate from Insulin $\left[\frac{\text{dL}}{\text{mIU} \cdot \text{min}}\right]$	
b_1	Insulin production in response to Glucose rate $\left[\frac{\text{dL}}{\text{mmg} \cdot \text{min}}\right]$	
b_2	Insulin removal in response to Insulin levels $\left[\frac{1}{\text{min}}\right]$	
τ	Glucose absorption delay [min]	
F	Total food consumed [mg]	

3 Results

3.1 Equilibrium analysis

Setting the changes of the dependent states to 0 and solving for G and I gives

$$0 = f(t) - a_1G - a_2GI \quad 0 = b_1G - b_2I$$

$$I = \frac{1}{2} \pm \sqrt{\frac{1}{4} + \frac{b_1f(t)}{a_1b_2}} \quad G = \frac{b_2}{2b_1} \pm \sqrt{\left(\frac{b_2}{2b_1}\right)^2 + \frac{b_2f(t)}{a_1b_1}}.$$

When no food is actively being digested, i.e. $f(t) = 0$, then

$$(G, I) = \left(\frac{b_2}{b_1}, 1\right), (0, 0).$$

These equilibrium points correspond to resting blood sugar levels and a degenerate system, respectively. To find the euilibrium of the non-degenerate solution we find the Jacobian of the system to get

$$J = \begin{bmatrix} \frac{d}{dG} \left(\frac{dG}{dt} \right) & \frac{d}{dI} \left(\frac{dG}{dt} \right) \\ \frac{d}{dG} \left(\frac{dI}{dt} \right) & \frac{d}{dI} \left(\frac{dI}{dt} \right) \end{bmatrix} = \begin{bmatrix} -a_1 - a_2I & -a_2G \\ b_1 & b_2 \end{bmatrix}.$$

The trace and determinate of the system are

$$\beta = -a_1 - a_2I + b_2 \quad \gamma = -a_1b_2 - a_2b_2I + a_2b_1G$$

respectivly. At the non-degernete steady state the trace and determinate give

$$\beta = -a_1 - a_2 + b_2 \quad \gamma = -a_1b_2 - 2a_2b_2$$

Which is stable sink when $b_2 < a_1 + a_2$ and $a_2 < -\frac{a_1}{2}$.

3.2 Simulations

The dataset this model was fit to from was HUPA0019P from the HUPA database. [3] This models parameters were optimized to the R squared of the glucose data. The model was fit to both after the patient ate a meal and when the patient was fasting. For when the patient was fasting $f(t)$ was set to 0. The model was solved using MATLAB's built-in 45 solver. Simulations were also plot using MATLAB.

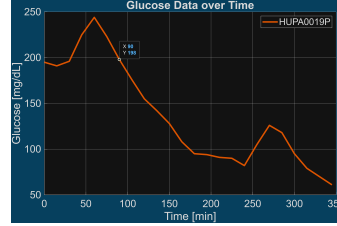
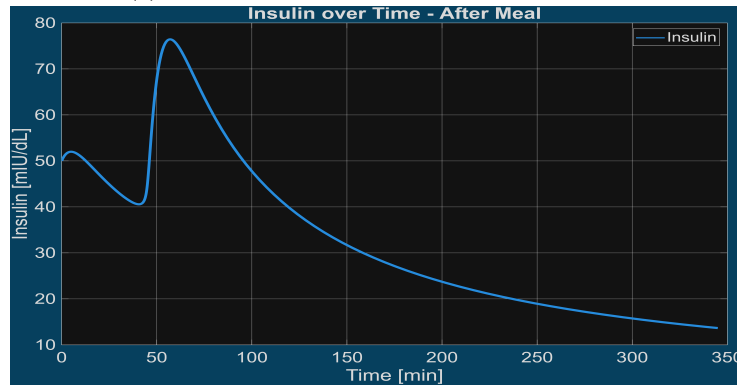


Figure 3: HUPA 0019P



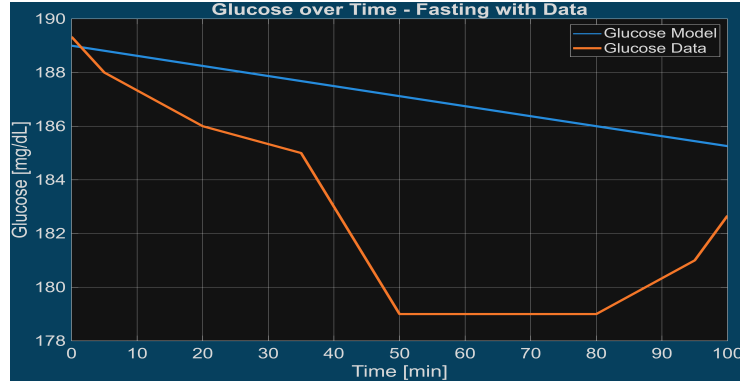
(a) Glucose over Time - After Meal with Data



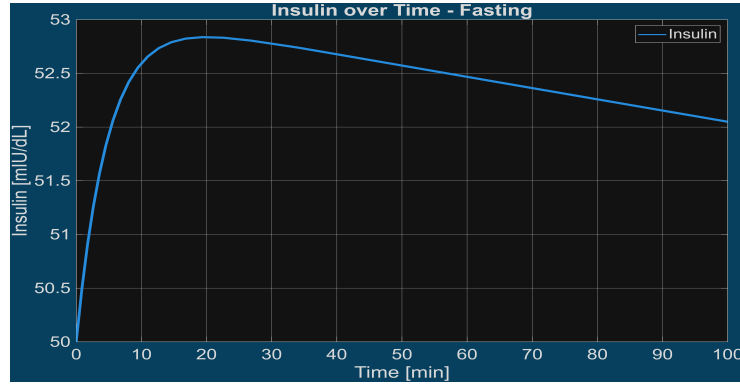
(b) Insulin over Time - After Meal

Figure 4: Fit Model - After Meal

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(a) Glucose over Time - Fasting with Data



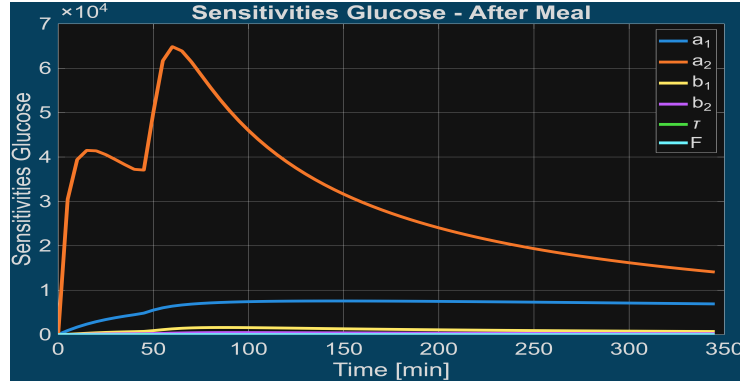
(b) Insulin over Time - Fasting

Figure 5: Fit Model - After Meal

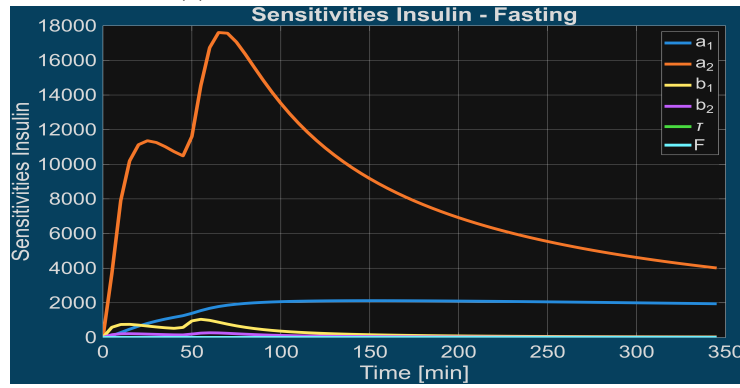
After Meal: The spike of the model is attributed to the glucose density function. The Data has a second spike which could be attributed to the patient eating again. The model shows that over time the body slowly consumes and stores the glucose to a stable state. The insulin spike after the meal can be described as the body's response to the increasing glucose levels. Once the glucose was back to a stable level the insulin asymptotes to a stable state as described by the equilibrium solution.

Fasting: The glucose slowly falls to a lower steady state but stays within a small range as would happen when the body is replenishing glucose levels by varying a_2 . Once the insulin rises to a steady level the small changes in insulin show that the body keeps the glucose consumption to a steady level. The average steady level of insulin of a healthy adult is approximately $17 \frac{\text{mIU}}{\text{dL}}$ which is significantly lower than the modeled steady state. This can be attributed to the patient's type 2 diabetes.

3.3 Sensitivity Analysis



(a) Sensitivities Glucose - After Meal

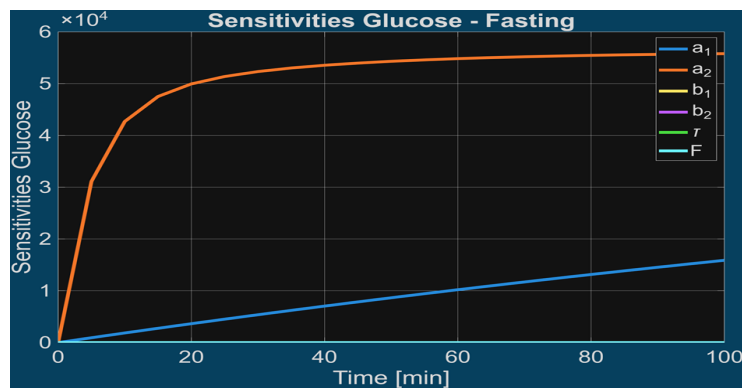


(b) Sensitivities Insulin - After Meal

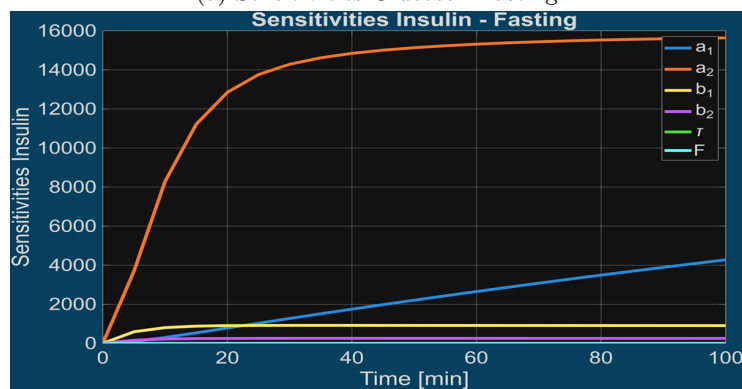
Figure 6: Sensitivities - After Meal

The model is strongly dependent on a_2 for both glucose and insulin levels after a meal. This can be explained by how responsive the cells of the body are to insulin and glucose levels. It can also be noted that towards the steady state a_1 is more sensitive as a_1 is defined by the body on longterm scales to keep glucose levels constant.

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(a) Sensitivities Glucose - Fasting



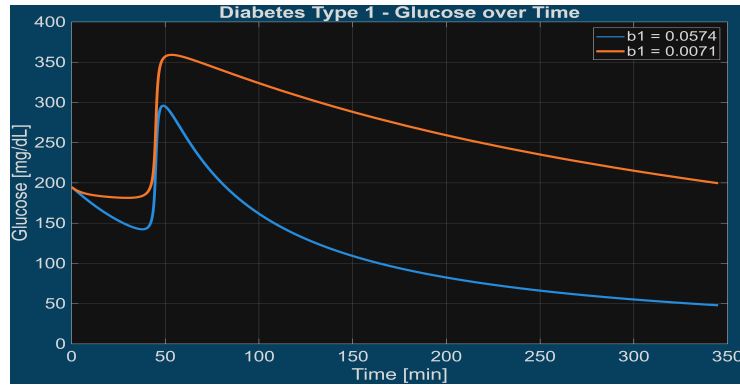
(b) Sensitivities Insulin - Fasting

Figure 7: Sensitivities - After Meal

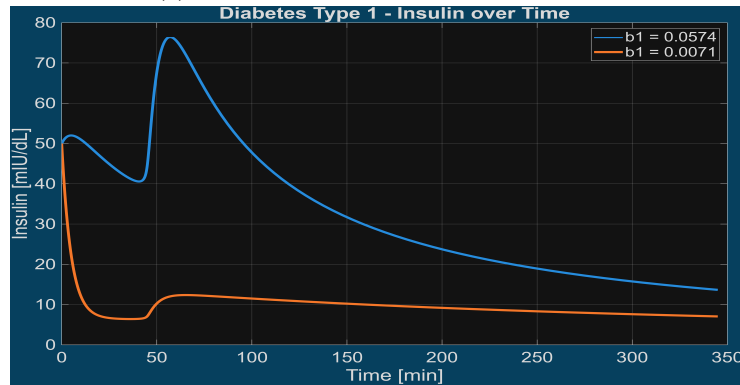
Similar to after a meal the a_2 is more sensitive at first but as time progresses a_1 is more important to manage glucose levels. b_1 which is the body's ability to produce insulin also had a role in the fasting steady state.

3.4 Diabetes Type 1

To model Diabetes Type 1 b_1 was lowered to describe how less β -cells would be able to produce Insulin.



(a) Diabetes Type 1 - Glucose over Time



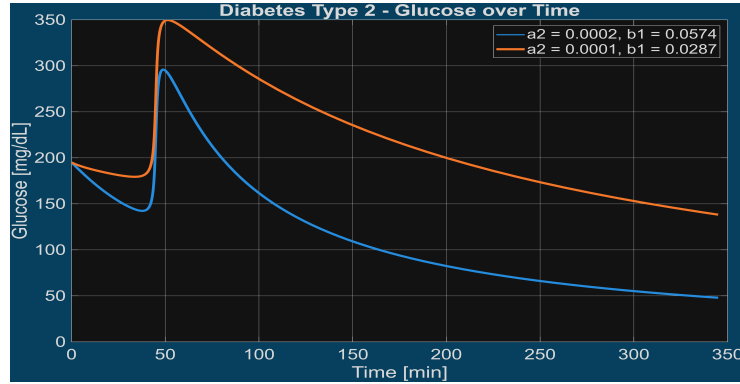
(b) Diabetes Type 1 - Insulin over Time.png

Figure 8: Diabetes Type 1

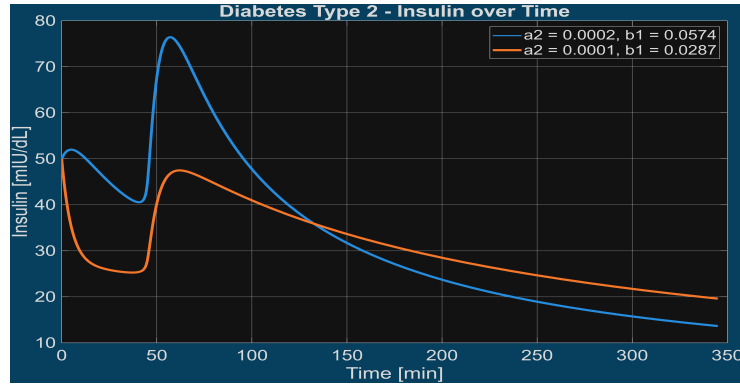
With type 1 diabetes the rise in glucose is both higher and stays elevated longer. As predicted insulin is produced at a lower rate.

3.5 Diabetes Type 2

To model Diabetes Type 1 b_1 was lowered to describe how less β -cells would be able to produce insulin to influence the cells of the body and a_2 was decreased to show that insulin had less of an influence over the body's cells.



(a) Diabetes Type 2 - Glucose over Time



(b) Diabetes Type 2 - Insulin over Time.png

Figure 9: Diabetes Type 2

Since insuling is produced at a slower rate then glucose is able to spike but differing to type 1 diabetes insulin is raised to a higher that average level to compinsate for the insulin resistance of the cells. This model also shows that monitoring insulin levels would be a more descriptive way to differentiate between the effects of simply diabetes and insulin resistance than just glucose levels.

4 Discussion

This model was difficult to optimize for food intake and time delay (F, τ) and may need to be adjusted for more accurate fitting of the data. Further work can be done to organize glucose intake amounts logged by the patient by time consumed, amount consumed, and general carb level. This model showed the interaction between the Blood Glucose and Insulin levels and could be explored for possible chaotic interactions. Bridgwater et al. explores a more complex statement of the model with bifurcation analysis.[1] The model accurately shows that glucose levels driven by insulin levels and shows that diabetes type 1 and type 2 differ by insulin effects on glucose levels.

4.1 Further Research

A further study could be done to study and diagnose insulin diseases by applying a statistical analysis to fitted models of larger datasets of continuous glucose monitoring. This model could be extended by attempting to define a_1 as a dependent compartment driven by metabolic intake.[4] This model could be adapted to account for the Glycogen storage for varying levels to correlate glycogen storage to long-term hormone levels.

5 Acknowledgments

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References

- [1] Adam Bridgewater, Benoit Huard, and Maia Angelova. “Amplitude and Frequency Variation in Nonlinear Glucose Dynamics with Multiple Delays via Periodic Perturbation”. In: *Journal of Nonlinear Science* 30.3 (June 2020), pp. 737–766. ISSN: 1432-1467. DOI: [10.1007/s00332-020-09612-1](https://doi.org/10.1007/s00332-020-09612-1). URL: <https://doi.org/10.1007/s00332-020-09612-1>.
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